

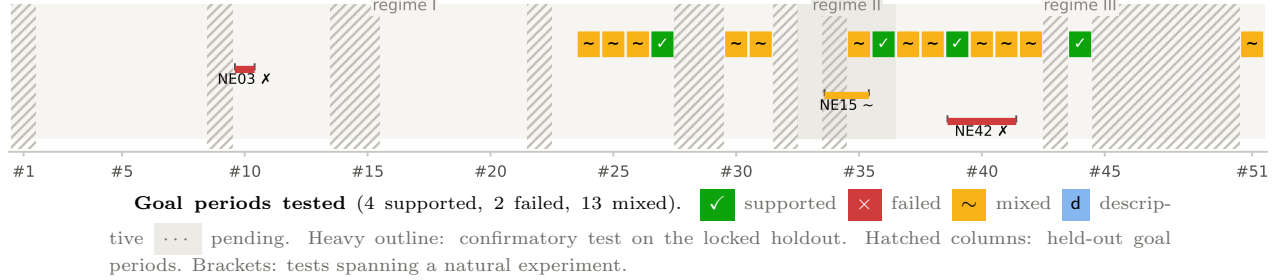
H18: Attention dilutes as 1/k: response to a message falls with the number of messages waiting at the agent's turn

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 48% Faithfulness 2.5/5 Usefulness 2/5

Bottom line. Attention in the village dilutes sub-linearly ($\hat{\beta} \approx 0.66$ on corrected visibility), giving coupling $\propto N^{-0.6}$ rather than a literal J/N ; replies go to the newest messages, and the placebo failure that clouded round 1 was mostly the visibility rule.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

At each talk turn we counted the k messages that reached the agent since its previous talk turn and asked whether it responds to each pending sender, fitting within agent-days in 16 non-holdout goal periods. Round 1b recounts k on the context ledger (which model call received which message) and adds reply labels. Response falls with k in 16/16 periods: pooled $\hat{\beta} = 0.66 \pm 0.02$ (round 1: 0.63). The curve saturates or follows recency; a literal $1/k$ wins 6/16. In #51, timer-wake batches of exogenous size give 0.45. Per-pair uptake falls with room size across periods ($\rho = -0.57$) and across #51's roster growth ($\rho \approx -0.8$). The round-1 placebo failure was mostly the visibility rule: invisible senders are now addressed at half the pending rate (was 85%), the rest is conversation state. Merge and fetch-limit tests failed. Holdout not run.

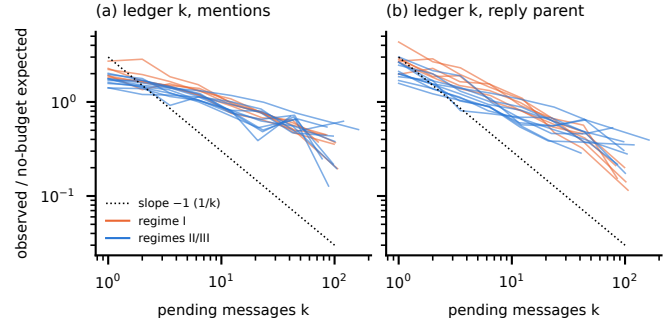
PHYSICAL MODEL

Mean-field coupling (model 01) with Hawkes-style per-pair cross-excitation (09). Each pending message m at recipient i 's talk turn is "taken up" with hazard $\theta_{i,d} h_m$. Sender j is responded to (named, or its message is the reply parent) if any of its messages is taken up. $\theta_{i,d}$: agent×day propensity. h : the attention rule (constant, $1/k$, saturating with k_0 , recency with ρ , power β). γ : boost for messages naming i . J_{ij} : per-pair coupling. N : room size. B : senders responded to per turn.

$$P(r_{\tau j} = 1) = 1 - \exp\left(-\theta_{i,d} \sum_{m \in P_j(\tau)} h_m\right)$$

$$h_m = e^{\gamma \mathbf{1}[m \rightarrow i]} \times \left\{1, \frac{1}{k}, \frac{1+k_0}{k_0+k}, \rho^{\text{rank}_m-1}, k^{-\beta}\right\}$$

$$J_{ij} \propto B k^{-\beta} \sim B N^{-\beta} \quad (\beta = 1: \text{literal } J/N)$$



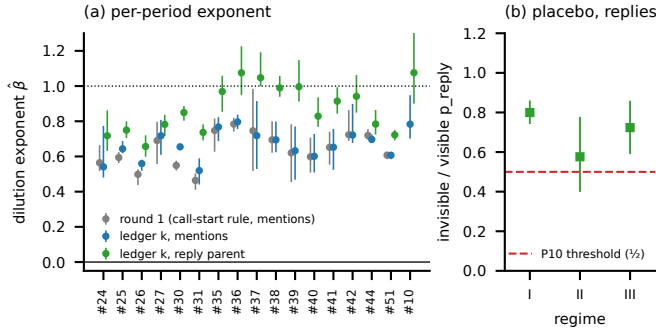
Observables. Response relative to a no-budget model with the same agent×day propensities, vs. pending k on the context ledger; one line per period. (a) Mentions: every period declines, more shallowly than $1/k$ (dotted). (b) Reply parents: steeper, close to $1/k$, because each message has one parent; which pending message gets it is decided mostly by recency.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For operators the usable fact is a scaling rule: in a shared channel, each agent's chance of engaging any one peer falls roughly as $N^{-0.6}$, while what it says per turn stays about the same. Bigger rooms get diluted, not more collaborative, and when an agent replies it replies to the newest messages. Splitting a channel raises within-room engagement by about $(k \text{ ratio})^{0.6}$, as at #35's split. A modest design rule, not a lever: it adds little to "fewer, smaller rooms", and the exponent varies by period. The mention signal is now cleaner (the placebo failure was mostly a visibility bug) but still carries conversation state.

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b (ledger k)	Verdict (1b)
P1 $\hat{\beta} > 0$, pooled 0.5–1.2	16/16, 0.63 → 16/16, 0.66 (replies 0.85)	yes
P2 budget shape beats recency	5/16 → 6/16; replies: recency 14/16	no
P4 mentions bypass dilution	1/16 → 1/16	no
P5 exogenous- k timer wakes	3/6 → 3/6; #51 0.50 → 0.45 [0.41, #51 yes 0.50]	yes
P6 room effect absorbed by k	3–4/8 → same	mixed
P7 05-04 (NE42 native) merge	failed → failed (raw); joint fit: k absorbs 53–86%	no
P8 regime invariance	Δ 0.14 → 0.07	yes
P9 $\bar{p}$ falls with N , $\hat{B}$ flat	ρ $-0.55/-0.03$ → $-0.57/-0.02$	yes
#51 N sweep (native) replies:	$\bar{p}$ ρ -0.80 , $\hat{B}$ -0.64	mixed
P10 invisible placebo	0/16, $0.85\times$ pending → 3/16, mostly explained $0.49\times$	ex-



Second figure. (a) Exponent per period: round 1 (gray), ledger k with mentions (blue), ledger k with reply parents (green). The visibility fix barely moves it; replies push it toward 1 by construction. (b) Placebo on reply labels: invisible pairs keep 60–80% of a visible pair’s reply score (threshold 50%).

GOAL PERIODS IN DETAIL

G24 ~ 1b: mixed : ledger k , mention response, pre-registere...
 G25 ~ 1b: mixed : ledger k , mention response, pre-registere...
 G26 ~ 1b: mixed : ledger k , mention response, pre-registere...
 G27 ✓ 1b: supported : ledger k , mention response, pre-regis...
 G30 ~ 1b: mixed : ledger k , mention response, pre-registere...
 G31 ~ 1b: mixed : ledger k , mention response, pre-registere...
 G35 ~ 1b: mixed : ledger k , mention response, pre-registere...
 G36 ✓ 1b: supported : ledger k , mention response, pre-regis...
 G37 ~ 1b: mixed : ledger k , mention response, pre-registere...
 G38 ~ 1b: mixed : ledger k , mention response, pre-registere...

+9 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Mentions mix uptake with conversation state; reply labels allow one parent per message and keep a thread-membership component, so neither is a clean per-message uptake. Recency and budget are still inseparable where k is small. Timer-wake identification works only in #51. The cross-period size law is confounded with era; #51 is the only large room. No multiplicity correction; nothing confirmed on the holdout, whose script still builds round-1 inputs.

ROUND 2

- **Essence.** Attention is the scarce resource, and coupling is its allocation.
- **H18-R1** A salience law, uptake = $f(\text{mention, recency, novelty, sender status})$, is fit and predicts held-out uptake (E5).
- **H18-R2** Total attention per turn is conserved; re-test with reply-threading labels.
- **H18-R3** An optimal room size maximizes total useful uptake (HH111).